

Rear Suspension Redesign

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Abstract

Rear suspension systems play a critical role in maintaining vehicle stability, traction, and ride quality, particularly for autonomous vehicles operating on uneven terrain. They are designed to improve passenger ride comfort by absorbing impacts, maintaining tire contact with uneven terrain, and reducing the amount of road vibration transmitted to the chassis. The current rear suspension on the Autonomous Vehicle Lab's vehicle of study, Carl, is excessively stiff due to restrictive welded hinges and rigid suspension geometry, causing the vehicle to rely primarily on tire deformation rather than proper suspension movement to absorb impacts.

The objective of this project was to redesign Carl's rear suspension using an independent trailing-arm suspension system that improves suspension compliance, ride comfort, and off-road performance while maintaining structural integrity and compatibility with the existing vehicle frame. Three rear suspension concepts were evaluated using a weighted decision matrix based on suspension compliance, structural integrity, manufacturing complexity, cost, ease of integration, and maintenance.

Introduction

Autonomous vehicles rely on effective suspension systems to maintain stability, traction, and ride quality while operating over uneven terrain. A properly designed suspension system allows the wheels to follow changes in the ground surface while reducing the transmission of impacts and vibrations to the vehicle chassis. This improves vehicle handling and helps onboard sensors to capture clear data without the intrusion of violent movements.

This poster presents the redesign of the rear suspension system for the Autonomous Vehicle Lab's vehicle of study, *Carl*. Evaluation of the existing rear suspension revealed that the original design is excessively stiff due to restrictive welded hinges and rigid suspension geometry. As a result, the suspension provides little functional wheel travel and relies primarily on tire deformation to absorb impacts.

To address these limitations, an independent trailing-arm rear suspension system was developed to improve suspension compliance while maintaining structural integrity and compatibility with the existing chassis. The redesigned system incorporates a structural cross member, double-shear pivot brackets, independent trailing arms, coil springs, and hydraulic dampers that allow each rear wheel to move independently. The design aims to improve ride comfort, increase wheel travel, and ground clearance.

Discussion

The redesigned rear suspension replaces Carl's original rigid rear suspension with an independent trailing-arm system while retaining the existing four-wheel-drive drivetrain and hub-mounted rear motors. The suspension is built around a rectangular structural crossmember mounted between the rear frame rails, which serves as the primary structural member by providing rigid mounting locations for the suspension while distributing loads throughout the chassis. Beneath the crossmember, each rear wheel is supported by a boxed steel trailing arm that pivots on a double-shear bracket, allowing the wheels to move independently while maintaining structural rigidity under vehicle loading. This independent wheel motion reduces the transfer of impacts between the left and right sides of the vehicle, improving suspension compliance and overall vehicle stability when traveling over uneven terrain.

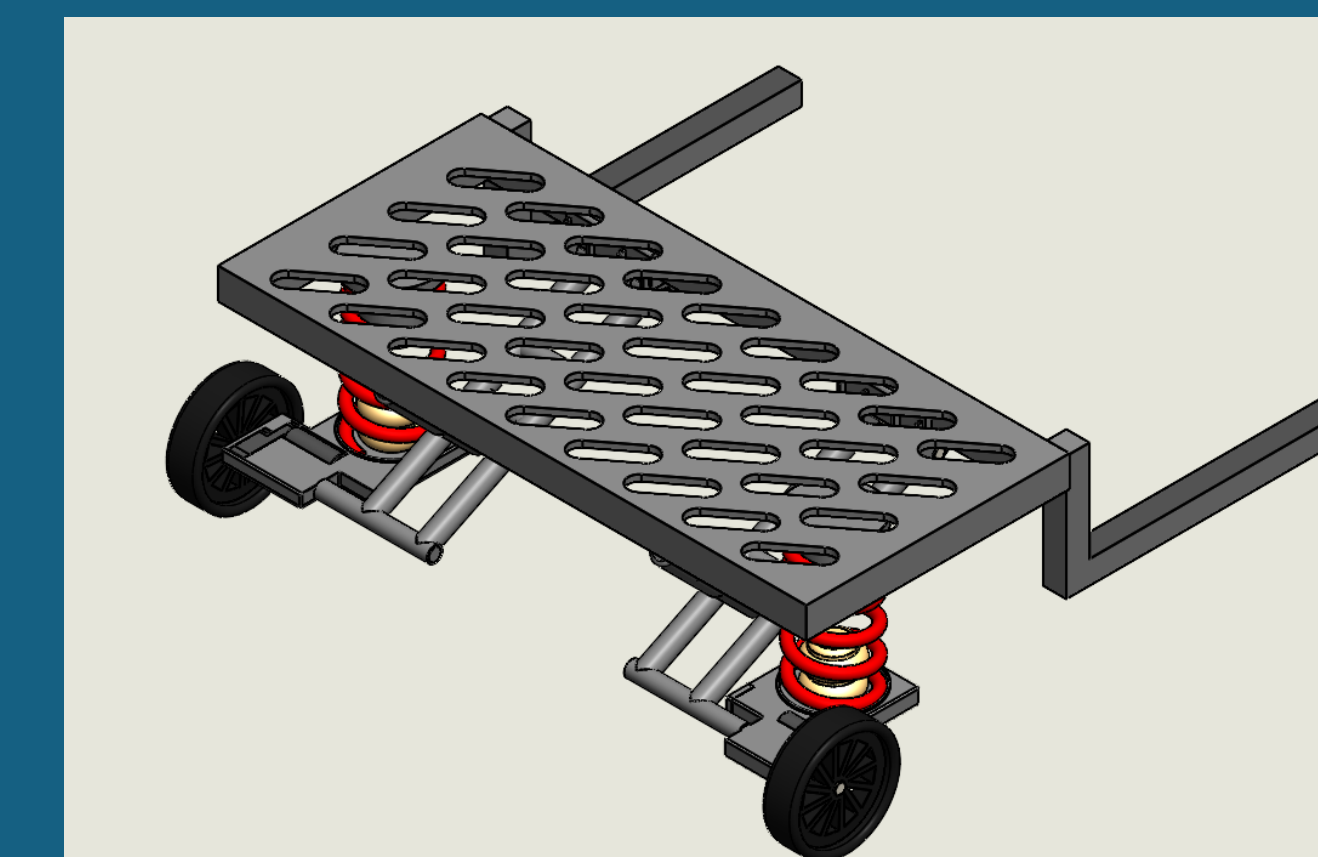
Each trailing arm supports the wheel assembly and existing electric drive motor, allowing the original drivetrain layout and electrical system to remain unchanged. Coil springs and hydraulic dampers mounted between the trailing arms and structural crossmember support the vehicle weight, absorb vertical impacts, and control suspension rebound to reduce chassis oscillations. Suspension loads are transferred from the tire contact patch through the wheel hub, trailing arm, spring, damper, and pivot bracket before being distributed into the structural crossmember and rear chassis. By eliminating the rigid axle connection found in the original suspension, the redesigned system provides greater suspension travel, improved ride comfort, more consistent tire contact with the ground, and better off-road performance while remaining compatible with the vehicle's existing frame, packaging constraints, and drivetrain configuration. The completed design provides a build-ready suspension package that can be manufactured and experimentally validated in future work.

Design Challenges

Rear Motor Integration: The existing rear motors had to remain securely mounted while allowing the new trailing-arm suspension to move freely. New motor mounts were designed to eliminate interference throughout suspension travel.



Packaging Constraints: The redesigned suspension had to fit within Carl's existing rear frame while remaining compatible with the drivetrain and body enclosure. Component placement was optimized to maximize suspension travel within the available space.



Weight Distribution: The added suspension components increased rear weight. The design was kept as lightweight as possible while maintaining the strength and rigidity needed for reliable suspension performance.

Conclusion

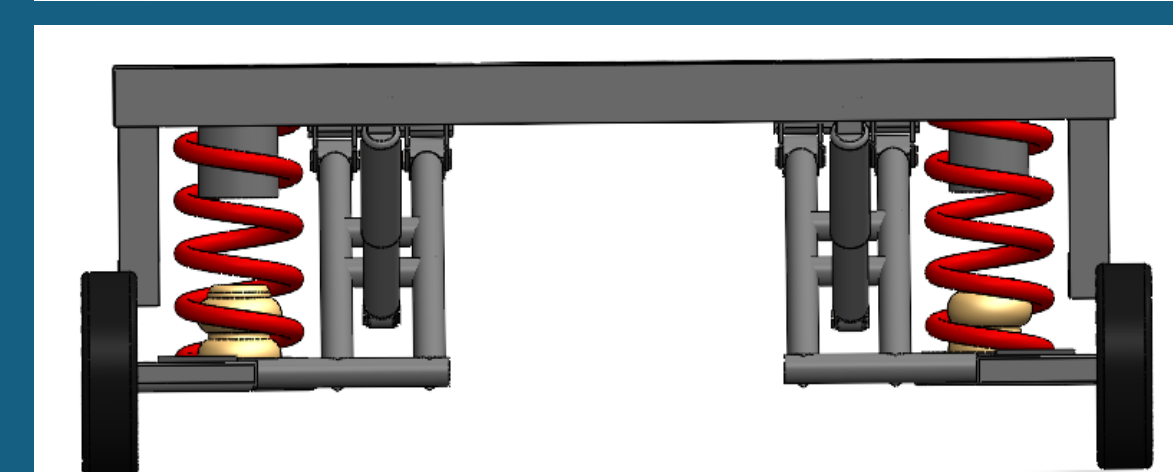
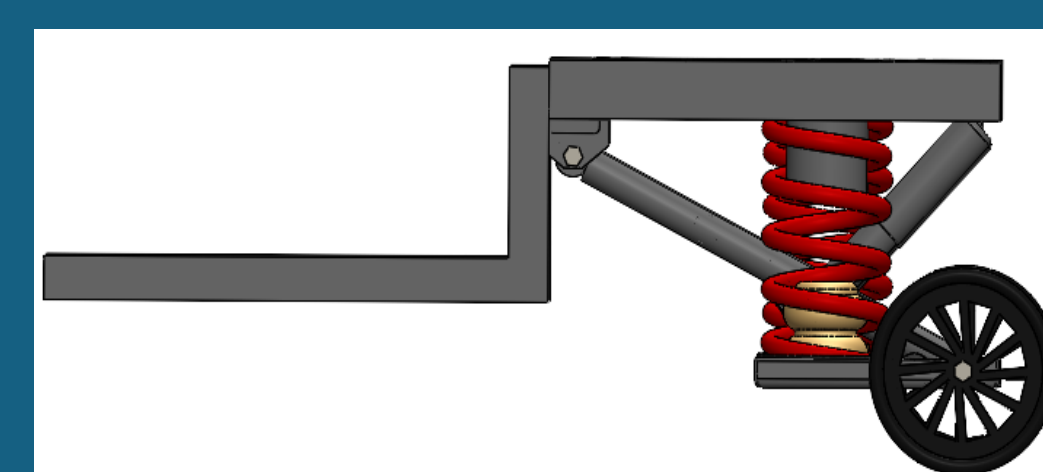
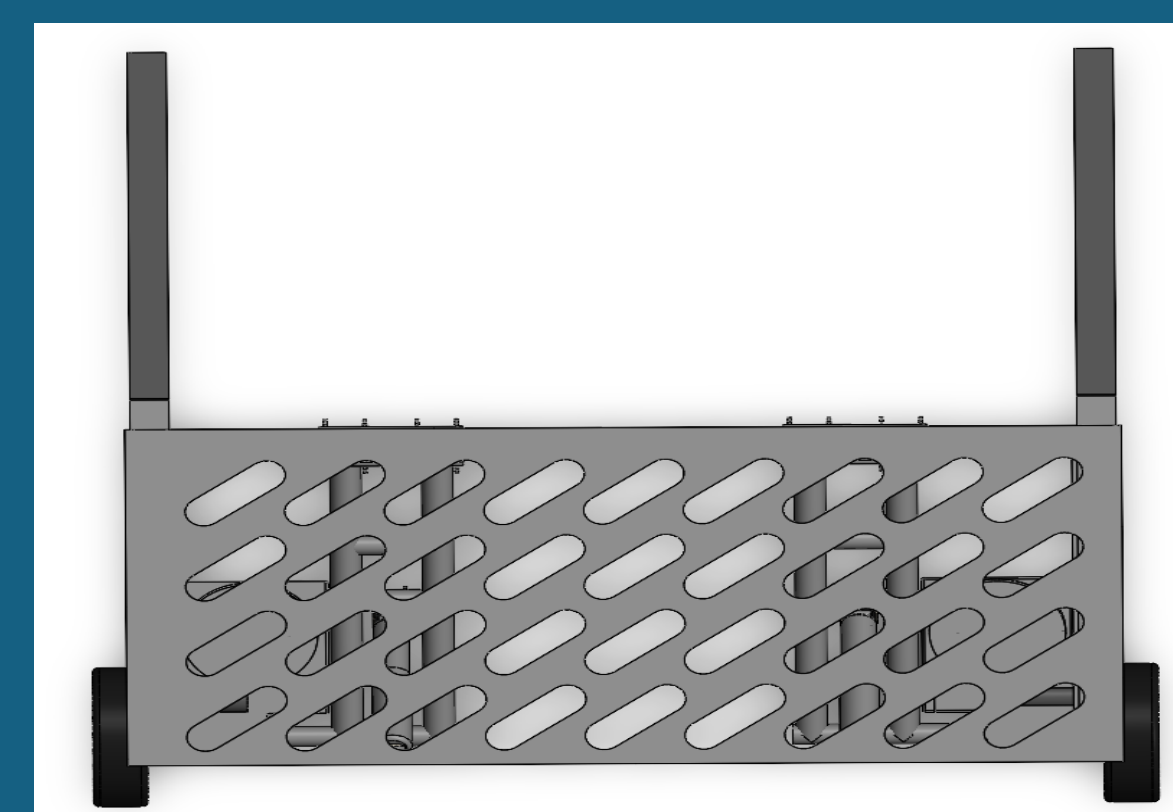
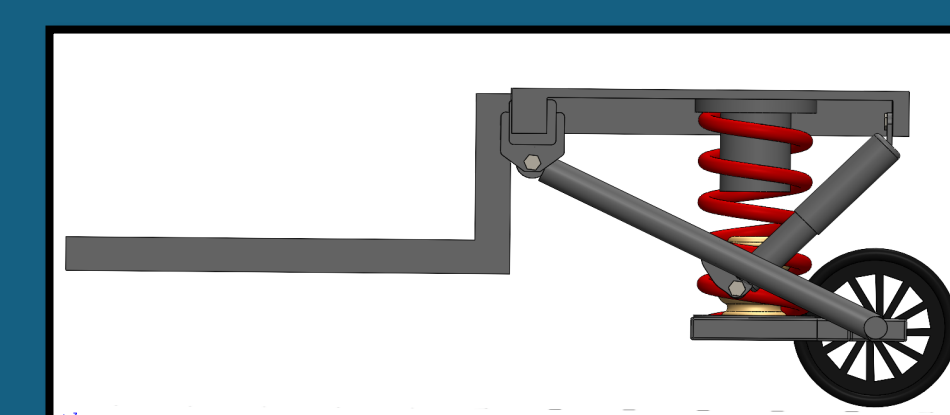
The redesigned rear suspension addresses the limitations of Carl's original rigid suspension by replacing it with an independent trailing-arm system that allows proper suspension movement while maintaining compatibility with the existing chassis and drivetrain.

The selected design incorporates a structural crossmember, trailing arms, pivot brackets, coil springs, and hydraulic dampers to improve suspension compliance, ride comfort, and off-road performance. Although fabrication and testing were outside the scope of this project, the completed design provides a foundation for future manufacturing and experimental validation, with the expected outcome of improved vehicle performance on uneven terrain.

Future Work

After the suspension has been installed, the vehicle should be tested over the same repeatable terrain course used during the baseline characterization. An IMU mounted to the chassis will again collect acceleration and vibration data, which will be processed in MATLAB using the same analysis procedure developed for the original suspension. Comparing the new results with the baseline data will allow the suspension's performance to be evaluated by measuring changes in oscillation frequency, damping characteristics, and overall ride quality.

3D CAD Model:



Reference

P. Try and M. Gebhard, "A Vibration Sensing Device Using a Six-Axis IMU and an Optimized Beam Structure for Activity Monitoring," *Sensors*, vol. 23, no. 19, pp. 8045–8045, Sep. 2023, doi: <https://doi.org/10.3390/s23198045>.

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